ECE 410 Codefest 3 CMAN

1. For 1 output element in C[i][j] All elements in B column j is accesed once. No other elements are accessed

total # accesed $= 2(32^3)$
$= 2(32768)$
$= 65536$

bytes $= 2(131072)$ bytes
$= 262144$ bytes

2. A traffic $= 4(32 \cdot 32)$
$= 4096$ bytes
B traffic $= 4(32 \cdot 32)$
total traffic $= 4096$ bytes

total traffic $= 8192$ bytes

3. $\frac{262144}{8192} = 32$

It equals N/T because the speed up is proportional to the matrix size N.

4. Naive

$\frac{262144 \text{ Bytes}}{320 \text{ GB/s}} = 0.8192 \mu s$

- memory bound

Tiled

$\frac{8192 \text{ bytes}}{320 \text{ GB/s}} = 0.0256 \mu s$

- memory bound